Problem 6 - Stochastic Gradient Descent

i) $P(\text{not choosing a sample}) = \frac{n-1}{n} = 1 - \frac{1}{n}$

with replacement $\Rightarrow$ n independent events

$$\Rightarrow \underbrace{\left(1-\frac{1}{n}\right) \times \cdots \times \left(1-\frac{1}{n}\right)}_{n \text{ times}}$$

$$\Rightarrow \boxed{P_{never} = \left(1-\frac{1}{n}\right)^n}$$

ii) $\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^x = e$; Show that $\lim_{n\to\infty} P_{never} \approx 0.3674$

$$\lim_{n\to\infty}\left(1-\frac{1}{n}\right)^n$$

let $y = \left(1-\frac{1}{n}\right)^n$

$$\ln(y) = n\ln\left(1-\frac{1}{n}\right)$$

$$\lim_{n\to\infty}\ln(y) = \lim_{n\to\infty}\frac{\ln\left(1-\frac{1}{n}\right)}{1/n} = \frac{\left(\frac{1}{1-\frac{1}{n}}\right)\left(\frac{1}{n^2}\right)}{-\frac{1}{n^2}} = -\frac{1}{1-\frac{1}{n}} = -1$$

$$\Rightarrow \lim_{n\to\infty}\ln(y) = -1$$

$$\Rightarrow \boxed{\lim_{n\to\infty} y = e^{-1} = 1/e \approx 0.3679}$$

iii) One random observation $Var(\nabla f(x^{(i)}, y^{(i)})) = \sigma^2$

Sampling with replacement $\Rightarrow$ random variables are i.i.d

for k iid random variables with variance $\sigma^2$

Sample mean $\bar{X} = \frac{1}{k}\sum_{i=1}^{k} X_i$

$$\boxed{Var(\bar{X}) = \frac{\sigma^2}{k}}$$

iv) The variance is reduced by a factor of k

v) [illegible] the batch size [illegible] may [illegible] noise [illegible] may want to reduce the learning [illegible] constant.

v) The claim in the paper is wrong because increasing the batch size and learning rate by a factor of k increases the variance by a factor of k

Let the variance of the gradient of one observation

$Var(\nabla f(x^{(i)}, y^{(i)})) = \sigma^2$

The variance of a mini batch of size k,

$Var(\nabla f(x, y)) = \frac{\sigma^2}{k}$ without changing the learning rate

The variance of a mini batch of size k and increasing the learning rate by a factor of k

$Var(k \nabla f(x, y)) = \frac{k^2 \sigma^2}{k} = k\sigma^2$

To maintain a constant variance $\sigma^2$, scale the batch size by k and scale the learning rate by $\sqrt{k}$.